Supplementary material

Table S1. ANOVA F-tests for the association between 24-hour movement behaviors and cognitive performance moderated by AD brain signatures

	Attentional/ inhibitory control		Episodic memory		Processing speed		Visuospatial processing		Working memory	
	F	p-value	F	p-value	F	p-value	F	p-value	F	p-value
Time-use composition	2.50	0.07	0.13	0.94	0.39	0.76	0.55	0.65	1.37	0.26
Time-use composition × GMMD signature	2.89	.041	5.16	.003	3.00	.036	2.65	.055	4.43	.006
Time-use composition × thickness/volume signature	0.42	.741	1.06	.369	0.11	.955	0.82	.485	1.07	.368

ANOVA F-test outputs from linear regression model comparisons investigating associations between overall 24-hour movement behavior composition and each cognitive domain moderated by Alzheimer's disease brain signatures. The 24-hour movement behavior composition was represented by three isometric log-ratio coordinates. Moderation was tested by comparing models with and without the three ILR × moderator interaction terms. GMMD and thickness/volume signatures were tested in separate models. Abbreviations: F, F statistic; GMMD, gray matter mean diffusivity; Attention/inhibition, attentional/inhibitory control.

#Table S2. Linear regression models showing the associations of movement behaviors with cognitive domains moderated by AD brain signatures.

	Thickness/volume signature			Gray matter mean diffusivity signature		
Behavior	n	β (95%CI)	p-value	n	β (95%CI)	p-value
Attentional/inhibitory control						
Moderate-to-Vigorous Physical Activity	89	0.03 (-0.16 to 0.23)	0.72	85	0.13 (-0.11 to 0.36)	0.29
Light Physical Activity	89	-0.3 (-1.09 to 0.48)	0.44	85	-0.71 (-1.38 to -0.03)	0.04
Sedentary Behavior	89	-0.06 (-0.44 to 0.32)	0.74	85	0.02 (-0.47 to 0.5)	0.95
Sleep	89	.005 (-0.47 to 0.48)	0.98	85	-0.24 (-0.77 to 0.28)	0.35
Episodic memory						
Moderate-to-Vigorous Physical Activity	89	-0.16 (-0.35 to 0.03)	0.10	85	0.32 (0.1 to 0.54)	.004
Light Physical Activity	89	0.19 (-0.59 to 0.96)	0.63	85	-0.89 (-1.54 to -0.23)	.008
Sedentary Behavior	89	0.29 (-0.07 to 0.66)	0.12	85	-0.39 (-0.86 to 0.08)	0.10
Sleep	89	0.42 (-0.04 to 0.89)	0.07	85	-0.66 (-1.16 to -0.17)	.009
Processing speed						
Moderate-to-Vigorous Physical Activity	89	-0.02 (-0.22 to 0.17)	0.82	85	0.1 (-0.13 to 0.34)	0.39
Light Physical Activity	89	0.22 (-0.56 to 0.99)	0.58	85	-0.71 (-1.39 to -0.03)	0.04
Sedentary Behavior	89	0.02 (-0.35 to 0.4)	0.90	85	0.07 (-0.41 to 0.56)	0.77
Sleep	89	0.02 (-0.46 to 0.49)	0.95	85	-0.19 (-0.71 to 0.33)	0.47
Visuospatial processing						
Moderate-to-Vigorous Physical Activity	89	-0.14 (-0.34 to 0.06)	0.16	85	0.1 (-0.13 to 0.33)	0.38
Light Physical Activity	89	0.05 (-0.76 to 0.85)	0.91	85	-0.13 (-0.81 to 0.55)	0.70
Sedentary Behavior	89	0.3 (-0.09 to 0.68)	0.13	85	-0.07 (-0.55 to 0.41)	0.78
Sleep	89	0.36 (-0.13 to 0.85)	0.14	85	-0.36 (-0.87 to 0.15)	0.16
Working memory						
Moderate-to-Vigorous Physical Activity	89	-0.11 (-0.3 to 0.08)	0.26	85	0.03 (-0.2 to 0.26)	0.80
Light Physical Activity	89	-0.22 (-1 to 0.56)	0.58	85	-0.77 (-1.42 to -0.12)	0.02
Sedentary Behavior	89	0.28 (-0.09 to 0.65)	0.14	85	0.27 (-0.2 to 0.74)	0.26
Sleep	89	0.29 (-0.18 to 0.76)	0.22	85	-0.02 (-0.53 to 0.49)	0.95

Independent models were constructed for each behavior, including the isometric log-ratio (ILR) coordinates of movement behaviors as predictors, and adjusted for sex and education. Results refer to the dominant behavior relative to the others. Abbreviations: 95%CI: 95% confidence interval. β: Unstandardized beta coefficients indicate how much the outcome changes for a one-unit increase in the respective ILR coordinate. This reflects a proportional increase in the dominant behavior relative to the rest of behaviors.

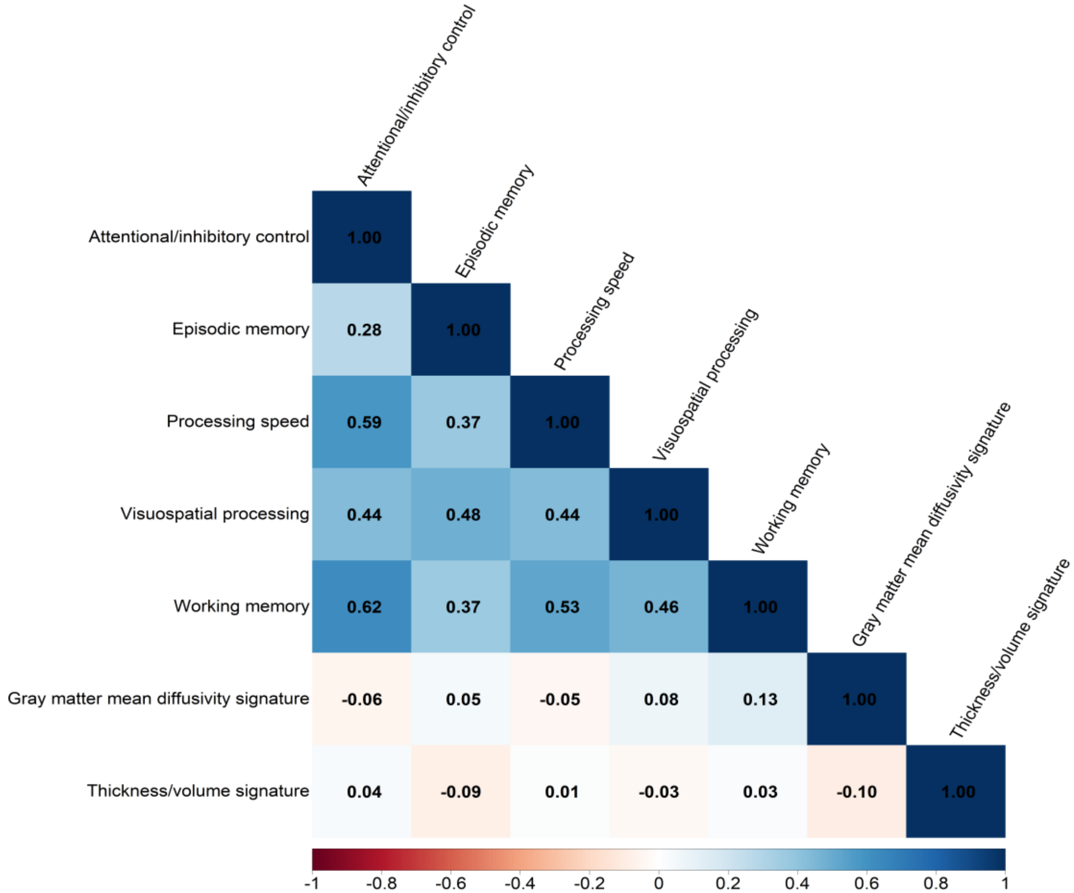

Figure S1. Pearson correlations between cognitive outcomes and AD signatures*. Color bar represents the strength of the correlation as r values. The darker blue the color, the stronger the positive correlation between variables, and the darker red the color, the stronger the negative correlation between variables. Higher values of the gray matter mean diffusivity (GMMD) signature indicate greater microstructural brain disruption (worse brain reserve), whereas higher cortical thickness/volume signature values indicate better structural integrity.*

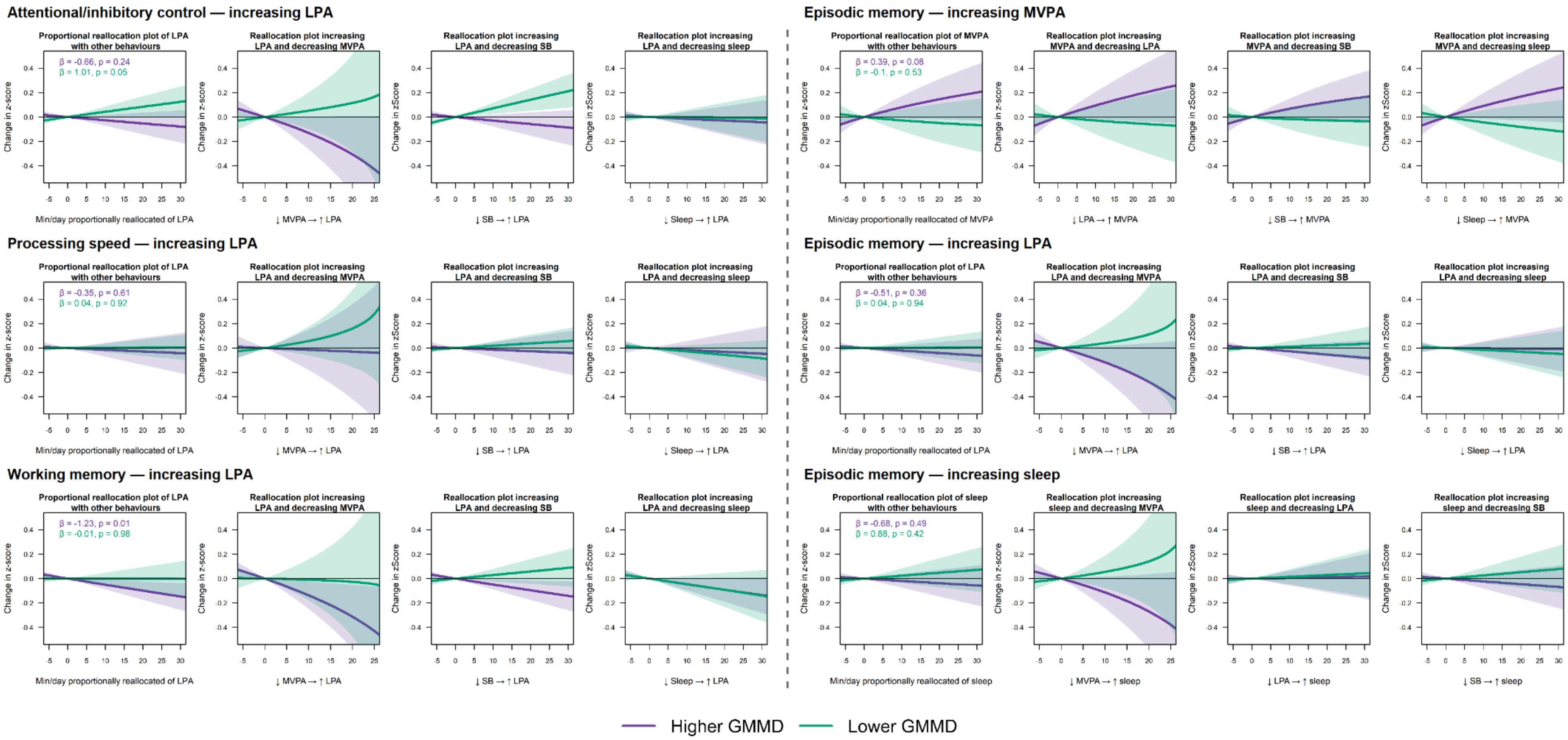

Figure S2. 1 Pairwise isotemporal reallocations of time between movement behaviors in relation to cognitive outcomes showing significant gray matter mean diffusivity (GMMD) moderation. The figure illustrates the estimated differences in cognitive outcomes associated with reallocating time (minutes/day) between movement behaviors — Moderate-to-Vigorous Physical Activity (MVPA), Light Physical Activity (LPA), Sedentary Behavior (SB), and Sleep , among individuals with lower and higher GMMD signature levels. Only cognitive domains with significant interaction are displayed.

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